Jack Willson

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github | linkedin

Personal Profile

Mechatronics Engineer (First Class Honours) with strong experience in vehicle dynamics, embedded systems, and electric vehicle control. Key contributor to the University of Canterbury Motorsport Formula SAE team, helping deliver an Overall EV Category win. Experienced in firmware, PCB design, CAN architecture, motorsport data analysis, and real-time control systems. I enjoy solving complex integration problems in high-performance engineering environments and am motivated by motorsport and embedded systems development.

Education

University of Canterbury — *2022-2025*

Bachelor of Engineering with Honours (Mechatronics) - First Class Honours

Motorsport Experience

University of Canterbury Motorsport — *May 2024-Dec 2025 / Christchurch*

Vehicle Dynamics & Electrical Engineer – Formula SAE Team

- Contributed to system-level design, development, and validation of the Formula SAE Electric vehicle, which won Overall EV in Australasia.
- Restructured the existing MoTeC M1 Build codebase for better modularity and integrated the new DTI inverters and motors.
- Implemented and validated PID-based traction control and integrated it with torque vectoring strategies.
- Designed the system-level CAN architecture and authored CAN DBC files for consistency across devices.
- Integrated and configured the Link PDM for power distribution, fault handling, and cooling control.
- Designed PCBs and housings for custom front wheel speed sensors and a custom driver interface with dials.

Professional Employment

Onfarm Solutions Ltd — *Nov 2024-Apr 2026 / Christchurch*

Junior Software Engineer

- Developed PLC logic and HMI interfaces with CODESYS (IEC 61131-3) for current Teatwand models and R&D projects.
- Integrated IO-Link devices and IoT systems through IO-Link masters and various industrial fieldbuses.

Tourism and Travel Ltd — *Nov 2023-Feb 2024 / Christchurch*

Mechanic

- Repairing and maintaining a fleet of over 150 vans including HiAce's, Mercedes Sprinters and LDV V80's.
- Complete interior and drivetrain checks of vans after every hire.

Triple S Engineering & Marine — *Mar 2022-Nov 2022 / Christchurch*

Machinist

- Operated manual mills and lathes to fabricate and assemble jet boat and marine components for export markets.

Wanaka Tractor Services — *2019-2023 / Wanaka*

Mechanic

- Maintained and repaired light and heavy vehicles, trailers, and agri-equipment, including engine and drivetrain work.
- Performed diagnostics and auto-electrical tasks to ensure reliable operation.

Skills

Programming	C/C++ | Python | Ladder Logic | IEC 61131-3 Structured Text | VHDL
Embedded & Control	CAN Bus/DBC | PID Control | ESP-IDF | Modbus/EtherCAT/Ethernet-IP | IO-Link
PCB & Design	Altium Designer | KiCad | SOLIDWORKS | Onshape
Motorsport & Automotive Tools	MoTeC M1 Build/Tune | MoTeC i2 | PDMLink | Wiring and Loom manufacture
Software & Workflow	Git | CI/CD | Linux | VS-Code | CODESYS

Projects

Wacky Races Embedded System Project *2025*

ARM Embedded Systems, 4-Layer PCB, RF Control

- Designed a remote-controlled vehicle and wearable motion controller using Microchip SAM4S ARM MCUs.
- Developed a 4-layer PCB (Altium) integrating a buck converter, ADXL345 Accelerometer, nRF24 RF, USB, and SWD.
- Implemented and managed firmware using VS-Code and Git with structured milestone-based hardware-software validation.

Fitness Monitor Software Restructure *2025*

Embedded Firmware Architecture

- Refactored embedded firmware into a modular layered architecture separating HAL, drivers, and application logic.
- Improved maintainability and scalability through structured unit testing and continuous integration Git workflows.

Go-Cart Controller *2025*

Power Electronics Project

- Designed, built, and tested a fixed-frequency current-mode controller for a 24 V DC motor in a go-cart, using the UC3843 PWM chip.
- Implemented overcurrent protection and accelerator control, bench-tested circuits, and optimised performance for and won the time-trial event.

Formula SAE Autonomous Track Guidance *2025*

Autonomous Vehicle Perception

- Developed a vision-based autonomous guidance system for a Formula Student car using a monocular camera and YOLOv8 CNN to detect track cones.
- Implemented midline estimation, polynomial trajectory smoothing, and filtering techniques to produce a smooth drivable path, achieving up to 95% track coverage in test videos.

Personal Linux File Server *2025*

Self-Hosted Server

- Set up and ran a Linux file server with network shares, and backups.
- Hosted Immich photo server for managing personal photos, keeping everything organised and accessible.

RoboCup Autonomous Robot Challenge *2024*

Autonomous Robotics, 3rd Place Overall

- Built a fully autonomous robot using a Teensy 4.0 (600 MHz Cortex-M7) for object collection in a dynamic arena.
- Developed real-time navigation, obstacle avoidance, and strategic task control under strict time and resource constraints.
- Designed mechanical collection systems and iteratively refined control algorithms through extensive testing.

Helicopter controller *2024*

C, CCS, Git, Tiva TM4C123G LaunchPad

- Programmed a microcontroller to monitor a helicopter's altitude and yaw using an ADC circular buffer and an interrupt-based quadrature encoder. The helicopter's motion was programmed using a PID controller.

Line Following Robot *2023*

C++, SOLIDWORKS, 3D-Printing, PCB Design, ATmega328

- Designed two PCB boards (microcontroller and sensors) and a 3D-printed chassis and drivetrain to build a robot. It was designed to follow a 20mm wide black line as fast as possible using a digital PID controller.

Extra

- University transcript is available on request.
- Referees are available on request.

August 6, 2026